

GATE 1

Measurement validity

frozen metric identity; repeatability within tolerance

GATE 2

Population and action support

held-out scenes; admitted targets; complete hard-valid tables

GATE 3

Actor-visible protocol

target matching; actor-input identity; leakage audit

GATE 4

Oracle headroom

equal-budget lookahead versus one-step oracle greedy

GATE 5

Actor-visible Q_1

scene-clustered ranking and calibration under admitted inputs

GATE 6

Exact Q_2

held-out recursive error and complete-support coverage

GATE 7

Endpoint gap closure

paired learned-myopic, finite-horizon learned, and oracle-lookahead endpoints

repeatability

supported

admissible

meaningful

learnable

recursive

Interpret the earliest failed prerequisite

Its own estimate remains visible; dependent claims are blocked, and missing evidence is never recorded as zero.